

Multi-Layer Perceptrons and Deep Learning

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Outline

Features and layers

- Introduction

- Layers

- Activation functions

Algorithms

- Random projection

- Back propagation

- Derivatives

- Cost functions

- Gradient descent

- Stochastic gradient descent in practice

Python libraries

- sklearn

- PyTorch

- TensorFlow

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Stochastic gradient descent in practice

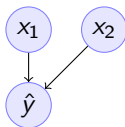
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Perceptron vs linear regression



- Network output

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- Chain rule

$$\nabla_{\beta} L = \nabla_{\hat{y}} L \nabla_{\beta} \hat{y}$$

- Network gradient

$$\nabla_{\beta} \hat{y} = (x_1, x_2)$$

Cost functions

The only difference are the cost functions

- Perceptron

$$L = -\mathbb{I}\{y \neq \hat{y}\} \hat{y}$$

with

$$\nabla L = -\mathbb{I}\{y \neq \hat{y}\} yx$$

- Linear regression

$$L = (\hat{y} - y)^2,$$

with

$$\nabla_{\hat{y}} L = 2(\hat{y} - y).$$

Layering and features

Fixed layers

- ▶ Input to layer $x \in R^n$
- ▶ Output from layer $\hat{y} \in R^m$.

Intermediate layers

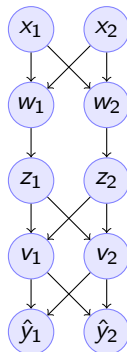
- ▶ Linear layer
- ▶ Non-linear **activation** function.

Linear layers types

- ▶ Dense
- ▶ Sparse
- ▶ Convolutional

Activation function

- ▶ Sigmoid
- ▶ Softmax



Input layer

Linear layer

Sigmoid activation

Linear layer

Softmax activation

Linear layers

Example: Linear regression with n inputs, m outputs.

- ▶ Input: Features $\mathbf{x} \in \mathbb{R}^n$
- ▶ Dense linear layer with $\Theta \in \mathbb{R}^{m \times n}$
- ▶ Output: $\hat{\mathbf{y}} \in \mathbb{R}^m$

Dense linear layer

- ▶ Parameters $\Theta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_m \end{pmatrix}$,
- ▶ $\theta_i = [\theta_{i,1}, \dots, \theta_{i,n}]$, θ_i connects the i -th output y_i to the features $\mathbf{x}$:
$$y_i = \theta_i \mathbf{x}$$
- ▶ In compact form:

$$\mathbf{y} = \Theta \mathbf{x}$$

Multiple linear layers

Repeated linear transformations are linear

It does not really help to have multiple linear layers one after the other. For example, if we transform $x \in \mathbb{R}^n$ to $z \in \mathbb{R}^k$ to $y \in \mathbb{R}^m$ through two matrices

$$z = Ax, \quad A \in \mathbb{R}^{k \times n} \quad (1)$$

$$y = Bz, \quad B \in \mathbb{R}^{m \times k} \quad (2)$$

We can rewrite y as

$$y = B(Ax) = (BA)x = Cx, \quad C \in \mathbb{R}^{m \times n} \quad (3)$$

where $C = BA$.

- ▶ Successive linear layers have no advantage normally.¹
- ▶ However, we can interlace them with **non-linear activation functions**.

¹Multi-task learning might be an exception.

ReLU activation

- ▶ Activation function:

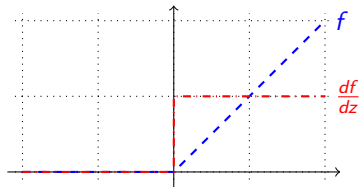
$$f(x) = \max(0, x)$$

- ▶ Derivative

$$\frac{d}{dx}f(x) = \mathbb{I}\{x > 0\}$$

Typically used in the hidden layers of neural networks, as it is:

- ▶ Simple to calculate.
- ▶ Nonlinear.
- ▶ Its gradient never vanishes.



Sigmoid activation

Example: Logistic regression

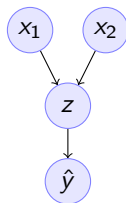
- ▶ Input $\mathbf{x} \in \mathbb{R}^n$
- ▶ Intermediate output: $z \in \mathbb{R}$,

$$z = \sum_{i=1}^n \theta_i x_i.$$

- ▶ Output: sigmoid activation
 $\hat{y} \in [0, 1]$.

$$f(z) = 1/[1 + \exp(-z)].$$

Now we can interpret $\hat{y} = P_{\theta}(y = 1|x)$.



Input layer

Linear layer

Sigmoid layer

Loss function: negative log likelihood

$$\ell(\hat{y}, y) = -[\mathbb{I}\{y = 1\} \ln(\hat{y}) + \mathbb{I}\{y = -1\} \ln(1 - \hat{y})]$$

Softmax layer

Example: Multivariate logistic regression with m classes.

- ▶ Input: **Features** $\mathbf{x} \in \mathbb{R}^n$
- ▶ Fully-connected **linear** activation layer

$$\mathbf{z} = \boldsymbol{\Theta} \mathbf{x}, \quad \boldsymbol{\Theta} \in \mathbb{R}^{m \times n}.$$

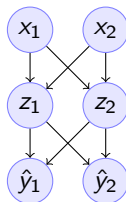
- ▶ **Softmax** output

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^m \exp(z_j)}$$

We can also interpret this as

$$\hat{y}_i \triangleq \mathbb{P}(y = i \mid \mathbf{x})$$

with usual loss $\ell(\hat{\mathbf{y}}, \mathbf{y}) = -\ln \hat{y}_y$



Input layer

Linear layer

Softmax layer

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Random projections

- ▶ Features x
- ▶ Hidden layer activation z
- ▶ Output y

Hidden layer: Random projection

Here we project the input into a high-dimensional space

$$z_i = \text{sgn}(\theta_i^\top x) = y_i$$

where $\Theta = [\theta_i]_{i=1}^m$, $\theta_{i,j} \sim \text{Normal}(0, 1)$

The reason for random projections

- ▶ The high dimension makes it easier to learn.
- ▶ The randomness ensures we are not learning something spurious.

Background on back-propagation

Gradient descent algorithm

- ▶ We need to minimise the expected value $\mathbb{E}_{\theta}[L]$ of the loss function L
- ▶ Since we cannot calculate $\mathbb{E}_{\theta}[L]$, we use:

$$\nabla_{\theta} \mathbb{E}_{\theta}[L] \approx \frac{1}{T} \sum_{t=1}^T \nabla_{\theta} \ell(x_t, y_t, \theta).$$

- ▶ We can then update our parameters to reduce the **empirical loss**

$$\theta_{t+1} = \theta_t - \alpha_t \nabla_{\theta} \ell(x_t, y_t, \theta).$$

The problem

- ▶ However ℓ is a complex function of θ .
- ▶ How can we obtain $\nabla_{\theta} \ell$?

The solution

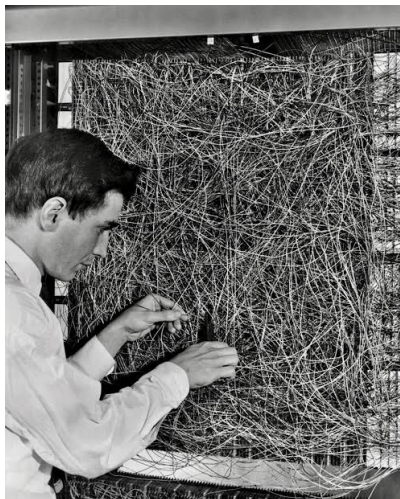
- ▶ Use the chain rule to "backpropagate" errors.

The chain rule of differentiation



[1673] Leibniz

Chain rule applied to the perceptron



[1976] Rosenblatt

Chain rule for deep neural networks



[1982] Werbos

Backpropagation given a name

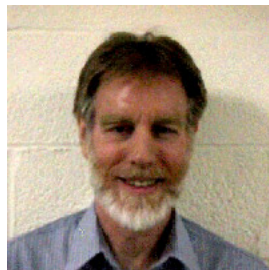
1986: Learning representations by back-propagating errors.



Rumelhart

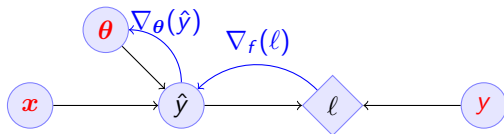


Hinton



Williams

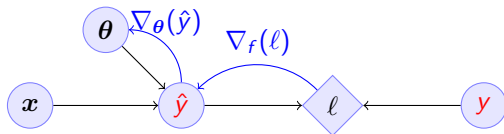
Elementary back-propagation: linear regression



- $f : X \rightarrow Y, \ell : Y \times Y \rightarrow \mathbb{R}$, chain rule: $\nabla_{\theta} \ell = \nabla_{\theta} f \nabla_{\hat{y}} \ell$
- **Forward:** follow the arrows to calculate **variables**

$$\hat{y} \triangleq f(\theta, x) = \sum_{i=1}^n \theta_i x_i, \quad \ell(\hat{y}, y) = (\hat{y} - y)^2$$

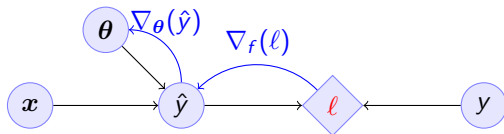
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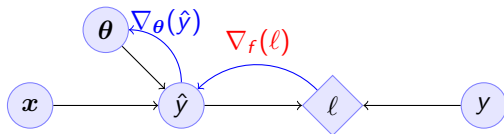
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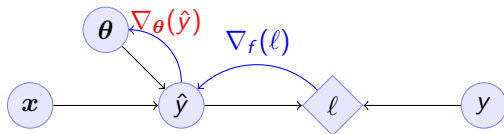
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► **Backward:** return to calculate the **gradients**

$$\nabla_{\theta} \ell(\hat{y}, y) = \nabla_{\theta} f(\theta, x) \times \nabla_{\hat{y}} \ell(\hat{y}, y) \quad (4)$$

$$= \nabla_{\theta} f(\theta, x) \times 2[\hat{y} - y] \quad (5)$$

Elementary back-propagation: linear regression



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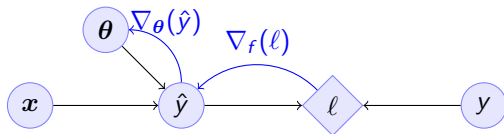
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► Update:

$$\theta_{t+1} = \theta_t - \alpha_t \times \nabla_{\theta} \ell(\hat{y}_t, y_t)$$

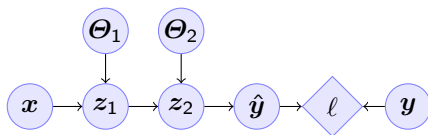
Gradient descent with *back-propagation*

- ▶ Dataset D , cost function $L = \sum_t \ell_t$
- ▶ Parameters $\Theta_1, \dots, \Theta_k$ with k layers
- ▶ Intermediate variables: $z_j = h_j(z_{j-1}, \Theta_j)$, $z_0 = x$, $z_k = \hat{y}$.

Gradient descent with *back-propagation*

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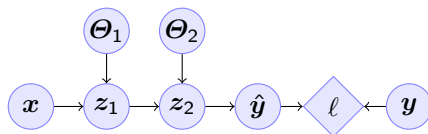
Dependency graph



Gradient descent with *back-propagation*

- ▶ Dataset D , cost function $L = \sum_t \ell_t$
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- ▶ Intermediate variables: $z_j = h_j(z_{j-1}, \Theta_j)$, $z_0 = x$, $z_k = \hat{y}$.

Dependency graph



Backpropagation with steepest stochastic gradient descent

- ▶ Forward step: For $j = 1, \dots, k$, calculate $z_j = h_j(k)$ and $\ell(\hat{y}, y)$
- ▶ Backward step: Calculate $\nabla_{\hat{y}} \ell$ and $d_j \triangleq \nabla_{\Theta_j} \ell = \nabla_{\Theta_j} z_j d_{j+1}$ for $j = k \dots, 1$
- ▶ Apply gradient: $\Theta_{j-} = \alpha d_j$.

Other algorithms and gradients

Natural gradient

Defined for probabilistic models

ADAM

Exponential moving average of gradient and square gradients

BFGS: Broyden–Fletcher–Goldfarb–Shanno algorithm

Newton-like method

Linear layer

Definition

This is a linear combination of inputs $x \in \mathbb{R}^n$ and parameter matrix $\Theta \in \mathbb{R}^{m \times n}$

$$\text{where } \Theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_i \\ \vdots \\ \theta_m \end{bmatrix} = \begin{bmatrix} \theta_{1,1} & \cdots & \theta_{1,j} & \cdots & \theta_{1,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \theta_{i,1} & \cdots & \theta_{i,j} & \cdots & \theta_{i,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \theta_{n,1} & \cdots & \theta_{n,j} & \cdots & \theta_{n,m} \end{bmatrix}$$

$$f(\Theta, x) = \Theta x \quad f_i(\Theta, x) = \theta_i \cdot x = \sum_{j=1}^n \theta_{i,j} x_j,$$

Gradient

Each partial derivative is simple:

$$\frac{\partial}{\partial \theta_{i,j}} f_k(\Theta, x) = \sum_{k=1}^n \frac{\partial}{\partial \theta_{i,j}} \theta_{i,k} x_k = x_j$$

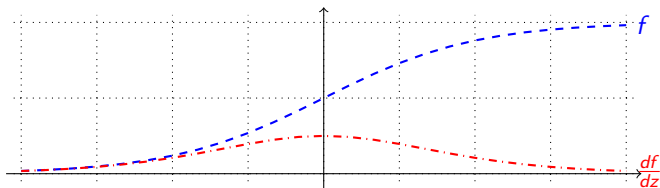
Sigmoid layer

- ▶ This layer is used for **binary classification**.
- ▶ It is used in the **logistic regression** model to obtain label probabilities.

$$f(z) = 1/(1 + \exp(-z))$$

- ▶ Derivative

$$\frac{d}{dz} f(z) = \exp(-z)/[1 + \exp(-z)]^2$$



Softmax layer

- ▶ This layer is used for **multi-class classification**
- ▶ It is a straightforward generalisation of the sigmoid function.

$$y_i(z) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

Derivative

$$\frac{\partial}{\partial z_i} y_i(z) = \frac{e^{z_i} e^{\sum_{j \neq i} z_j}}{\left(\sum_j e^{z_j}\right)^2}$$

$$\frac{\partial}{\partial z_i} y_k(z) = \frac{e^{z_i + z_k}}{\left(\sum_j e^{z_j}\right)^2}$$

Classification cost functions

Classification error

If z is the output for each class then

$$\ell(z, y) = \mathbb{I}\{y \notin \arg \max(z)\}$$

This is not differentiable.

Error margin

If z is the positive class output then

$$\ell(z, y) = -\mathbb{I}\{zy < 0\} zy$$

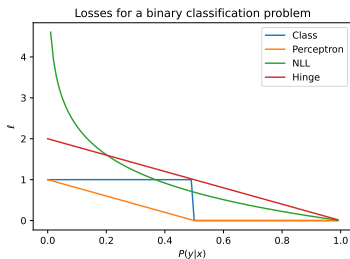
Used in the perceptron.

Negative log likelihood

If z are label probabilities, then

$$\ell(z, y) = -\ln z_y.$$

Used in logistic regression.



Hinge loss

If z are the output for each class

$$\ell(z, y) = 1 - z_y$$

Used in large margin classifiers.

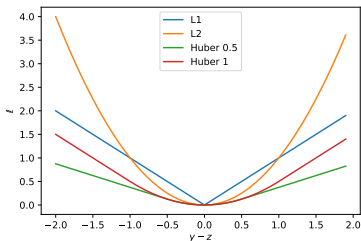
Regression cost functions

L2 loss (Squared error)

If z is a prediction for y then

$$\ell(z, y) = (y - z)^2$$

This is equivalent to negative log likelihood under Gaussianity. Used in linear regression.



L1 loss

If z is a prediction for y then

$$\ell(z, y) = |y - z|$$

Used in LASSO regression.

Huber loss

If z is a prediction, then

$$\ell(z, y) = \begin{cases} \frac{1}{2}(z - y)^2 & |z - y| \leq \delta \\ \delta(|z - y| - \frac{1}{2}\delta) & \text{otherwise.} \end{cases} \quad (6)$$

Mixes L1 and L2 losses.

Smooth function

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is ℓ -smooth if:

$$\|\nabla_x f(x) - \nabla_y f(y)\|_2 \leq \ell \|x - y\|_2.$$

Contraction mappings

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a contraction for some norm $\|\cdot\|$ if

$$\|f(x) - f(y)\| \leq \|x - y\|.$$

In other words, it is a contraction if it is 1-Lipschitz. In addition, contraction mappings have a fixed point x^* such that $f(x^*) = x^*$.

Gradient descent as a contraction

Suppose $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and ℓ -smooth, Then the mapping

$$\psi(x) \triangleq x - \eta \nabla_x f(x)$$

is a contraction as long as $\eta \leq 2/\ell$.

[See Nesterov 04 or Appendix A of Iterative Privacy Amplification for proofs]

Gradient descent in practice

The ideal gradient descent algorithm:

If we could calculate $\nabla_{\theta} \mathbb{E}_{\theta}[L]$, we could do:

$$\theta_{n+1} = \theta_n - \alpha_n \nabla_{\theta} \mathbb{E}_{\theta}[L]$$

for a suitable α_n schedule.

Gradient descent on the empirical error

Since we only have the data, we can try to minimise the empirical loss $\frac{1}{T} \sum_{t=1}^T \ell(x_t, y_t, \theta)$ through gradient descent

$$\theta_{n+1} = \theta_n - \alpha_n \frac{1}{T} \sum_{t=1}^T \nabla_{\theta} \ell(x_t, y_t, \theta)$$

This is also called **batch** gradient descent.

Stochastic gradient descent

Gradient descent on one example:

We don't have to wait calculate $\nabla_{\theta} \ell(x_t, y_t, \theta)$ for all t before applying the update. We can do it at every example:

$$\theta_{n+1} = \theta_n - \alpha_n \nabla_{\theta} \ell(x_{[n]_T}, y_{[n]_T}, \theta).$$

Here $[n]_T$ is $1 + n$ modulo T to ensure $n \in \{1, \dots, T\}$.

Minibatch gradient descent

However, it is a bit better to look at K examples at a time before we change the parameters. This is called a **minibatch**

$$\theta_{n+1} = \theta_n - \alpha_n \frac{1}{K} \sum_{k=nK}^{(n+1)K-1} \nabla_{\theta} \ell(x_{[k]_T}, y_{[k]_T}, \theta)$$

This also helps with parallelisation, since we can compute $\ell, \nabla_{\theta} \ell$ in parallel for each example.

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sklearn neural networks

Classification

Uses the **cross entropy** cost

```
from sklearn.neural_network import MLPClassifier
clf = MLPClassifier(hidden_layer_sizes=(5, 2))
clf.fit(X, y)
clf.predict(X_test)
```

- ▶ Main condition is layer sizes.

Regression

```
from sklearn.neural_network import MLPRegressor
model = MLPRegressor(hidden_layer_sizes=(5, 2))
```

PyTorch

Data set-up

```
X_train = torch.tensor(X_train, dtype=torch.float32)
train_dataset = TensorDataset(X_train, y_train)
train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
```

PyTorch: Manual training

Network setup

```
fc1 = nn.Linear(input_size, hidden_size) # Input to hidden layer
fc2 = nn.Linear(hidden_size, output_size) # Hidden layer to output
sigmoid = nn.Sigmoid() # some activation function
criterion = nn.BCELoss() #what loss to minimise
optimizer = optim.SGD(model.parameters(), lr=0.001) # how to minimise
```

Training

```
# Manual forward pass.
z1 = fc1(inputs) # hidden layer 1
a1 = sigmoid(z1) # Apply activation for hidden
z2 = fc2(a1) # Linear combination in output layer
outputs = sigmoid(z2) # Output layer activation
loss = criterion(outputs, labels) # Specify loss
loss.backward() # Backward pass
optimizer.step() # Update weights
```

TensorFlow

This is another library, no need to use this for this course a